SmartTrack - Detailed Hardware Connections and Verification Manu

Project: SmartTrack GPS-based smart tracking and emergency response system
Target MCU: ESP32-S3 N16R8 (16 MB flash / 8 MB OPI PSRAM)
Firmware framework: Arduino core 3.x / PlatformIO target
Document status: Engineering connection reference; bench verification is still required
Authoritative source: supplied CONNECTIONS_PER_COMPONENT.pdf and the project pin contract
Reference PDF SHA-256: dd20b8c0fef6298299574673f23cc62609d07a380c66f4268f2f78ec37f64538

1. How to use this document

This manual describes the wiring contract supplied for the SmartTrack build. It is intended to be used during schematic review, harness construction, firmware configuration, continuity testing, and staged bring-up.

The GPIO numbers below are authoritative for this build. Do not substitute a different GPIO because it is convenient in a sketch. Any change must update the schematic, firmware configuration, test plan, and this manual together.

Evidence labels

- **Reference-defined:** stated in the supplied connection document.
- **Firmware-defined:** represented in my projectt/firmware/.
- **Bench-required:** cannot be proven from source files; requires the assembled board and a meter, oscilloscope, or module test.
- **Reserved/unwired:** intentionally not connected in the current build.

Golden electrical rule

All module logic is **3.3 V only**. Never feed 5 V into an ESP32-S3 GPIO, SPI line, I2C line, UART line, button input, or sensor strap. The ordinary 5 V in the reference is USB VBUS before the board's onboard 3V3 regulator; it is not an ESP32 logic rail.

2. Complete net and rail summary

```
Net / function ESP32-S3 GPIO Direction from MCU Rail Connected component(s) State / note
SPI SCK 12 OUT 3V3 switched ILI9341 + XPT2046 Shared master clock
SPI MOSI 11 OUT 3V3 switched ILI9341 + XPT2046 MCU drives data to slaves
SPI MISO 13 IN 3V3 switched ILI9341 + XPT2046 Only selected CS-low slave drives
TFT chip select 10 OUT 3V3 switched ILI9341 Active LOW
TFT data/command 9 OUT 3V3 switched ILI9341 DC/RS
TFT reset 8 OUT 3V3 switched ILI9341 Active LOW
TFT backlight control 5 OUT 3V3 switched NPN driver PWM; never drive LED current directly
Touch chip select 14 OUT 3V3 switched XPT2046 Active LOW
Touch IRQ 15 IN 3V3 switched XPT2046 Wired; firmware may poll
I2C SDA 17 I/O 3V3 switched MPU-9250 4.7 kohm pull-up to 3V3
I2C SCL 18 OUT 3V3 switched MPU-9250 4.7 kohm pull-up to 3V3
MPU data-ready INT 16 IN 3V3 switched MPU-9250 Reference-defined; current firmware gap
GPS UART RX 6 IN 3V3 switched GPS TX UART1 crossover; 9600 baud
GPS UART TX 7 OUT 3V3 switched GPS RX UART1 crossover; 9600 baud
Power button 4 IN 3V3 always-on Button to GND External 10 kohm pull-up; RTC-capable
SOS button 1 IN 3V3 always-on Button to GND Internal pull-up; hold >=1200 ms
User button 2 IN 3V3 always-on Button to GND Internal pull-up; long press >=2000 ms
Peripheral enable 21 OUT / hi-Z always-on gate P-FET high-side switch LOW = switched rail ON
Modem RX reservation 48 reserved separate modem A7670C future UART Leave unconnected now
Modem TX reservation 47 reserved separate modem A7670C future UART Leave unconnected now
microSD CS -1 disabled not assigned microSD future Do not guess a GPIO
Modem PWRKEY -1 disabled separate modem A7670C future No current ESP32 assignment
```

```
Common ground GND power return GND Every module/button/MCU One common ground is mandatory
```

3. Shared-bus wiring

3.1 Shared SPI bus

Wire the SPI bus once, then fan it out to the TFT and touch controller:

```
ESP32 GPIO12 (SCK) --------- ILI9341 SCK / XPT2046 T_CLK
ESP32 GPIO11 (MOSI) --------- ILI9341 SDI / XPT2046 T_DIN
ESP32 GPIO13 (MISO) --------- ILI9341 SDO / XPT2046 T_DO
ESP32 GPIO10 (CS) --------- ILI9341 CS
ESP32 GPIO14 (CS) --------- XPT2046 T_CS
ESP32 GPIO15 --------- XPT2046 T_IRQ
```

The ESP32 is the only SPI master. SCK and MOSI are always MCU-driven. MISO is shared and depends on the unselected device releasing the line to high impedance.

Mandatory CS rule: GPIO10 and GPIO14 must never be LOW at the same instant. Before selecting one device, drive the other device's CS HIGH. If a future microSD device is added, its qualified CS must also be mutually exclusive with both GPIO10 and GPIO14.

At boot, de-select all SPI slaves before enabling the switched rail or starting a transaction. Use explicit SPI transactions when device modes or clock speeds differ. The current firmware declares the pins, but the TFT/touch runtime driver and complete CS arbitration still require integration and bench testing.

3.2 I2C bus

```
ESP32 GPIO17 (SDA) ----- MPU-9250 SDA
ESP32 GPIO18 (SCL) ----- MPU-9250 SCL
ESP32 GPIO16 (INT) ----- MPU-9250 INT (reference-defined)
ESP32 GND ----- MPU-9250 GND
3V3 switched ----- MPU-9250 VCC/VDD
```

Use 4.7 kohm pull-ups from SDA and SCL to 3.3 V. Verify whether the breakout already contains pull-ups; do not blindly install duplicate pull-ups. SDA/SCL must not be connected to 5 V.

4. ILI9341 TFT display

```
ILI9341 module pin Connect to Rail / direction Detailed note
VCC switched 3V3 power Turns off with GPIO21 rail
GND common GND return Required shared ground
SCK/CLK GPIO12 OUT Shared SPI clock
SDI/MOSI GPIO11 OUT Shared SPI data-in
SDO/MISO GPIO13 IN Must release when CS HIGH
CS GPIO10 OUT Active LOW; TFT-only
DC/RS GPIO9 OUT Data/command select
RESET GPIO8 OUT Active LOW
LED/BLK GPIO5 through NPN + base resistor OUT/PWM Do not source panel current from GPIO
```

Typical reference load is about 20 mA for logic plus about 80 mA for the backlight at full brightness. The backlight may be PWM-dimmed. The exact display breakout's regulator, LED topology, and transistor values remain **bench/BOM verification items**.

TFT bring-up

1. Confirm VCC is 3.3 V switched and GND is common.
2. Confirm RESET, DC, CS, and shared SPI lines with continuity testing.
3. Keep touch CS HIGH during TFT initialization.
4. Reset the panel LOW then HIGH according to the selected library.

5. Enable the NPN backlight driver only after the rail and logic are confirmed.
6. Run a fill-color and text test before adding touch transactions.

White or blank TFT symptoms commonly indicate RESET GPIO8, DC GPIO9, CS GPIO10, backlight GPIO5, or power-rail wiring faults.

5. XPT2046 touch controller

```
XPT2046 module pin Connect to Note
VCC switched 3V3 Less than 1 mA typical reference load
GND common GND Shared return
T_CLK GPIO12 Shared SPI SCK
T_DIN GPIO11 Shared SPI MOSI
T_DO GPIO13 Shared SPI MISO; high-Z when CS HIGH
T_CS GPIO14 Active LOW; touch-only
T_IRQ GPIO15 Wired input; current reference says firmware polls it
```

The touch controller must not share a LOW CS interval with the TFT. Restore each device's SPI mode and speed if the selected libraries require different settings. The current firmware has the pin contract but does not yet prove an integrated XPT2046 driver, calibration flow, or runtime CS lock.

6. u-blox NEO-M8N GPS

```
GPS pin ESP32-S3 connection Detail
VCC switched 3V3 Approximately 29 mA typical, 67 mA maximum reference
GND common GND Required
GPS TX (output) GPIO6 / ESP RX UART1 crossover
GPS RX (input) GPIO7 / ESP TX UART1 crossover
ANT active antenna Clear sky view; away from switching noise
```

Use UART1 at 9600 baud. The wiring must be crossed: **GPS TX -> ESP GPIO6 RX**, and **ESP GPIO7 TX -> GPS RX**. Never wire TX-to-TX or RX-to-RX.

Place approximately 100 uF bulk capacitance close to GPS VCC and another bulk capacitor on the switched rail near the GPS. The reference gives roughly 50 mA cold-acquisition current and another 10-20 mA for an active antenna. Cold fix time can be approximately 30 seconds to several minutes depending on antenna and sky view.

Keep the antenna away from the ESP32, TFT, switching supply, high-current modem wiring, motors, speakers, and metal enclosure parts. GPS bytes without a fix usually indicate antenna/sky-view or acquisition conditions; no bytes usually indicate crossover, baud, VCC, or switched-rail faults.

7. MPU-9250 IMU

```
MPU-9250 pin Connect to Detail
VCC/VDD switched 3V3 3.3 V only
GND common GND Required
SDA GPIO17 I2C bidirectional
SCL GPIO18 I2C clock
INT GPIO16 Data-ready input; firmware integration gap
AD0 GND Selects address 0x68
NCS/CS 3V3 switched Forces I2C mode; do not float
FSYNC GND Disables frame sync
EDA/ECL unconnected Auxiliary external-magnetometer bus unused
```

The expected address is 0x68. Verify WHO_AM_I on the actual breakout. The reference mentions approximately 3.9 mA active and very low sleep current. Keep the IMU away from magnets, motors, speakers, steel hardware, and high-current modem paths. Verify breakout pull-ups and board

orientation before calibration.

Current project gap: the firmware initializes Wire but leaves MPU acquisition/calibration unavailable, and the current configuration does not yet represent GPIO16 or the static strap wiring in code. The connection is documented, not fully software-verified.

8. Buttons and always-on controls

Power button - GPIO4

Connect one side of the button to GPIO4 and the other side to common GND. Use an **external 10 kohm pull-up to always-on 3.3 V** and a 100 nF debounce capacitor from GPIO4 to GND. It is active LOW, RTC-capable, and intended for long-press shutdown and deep-sleep wake.

Do not put this pull-up on the switched rail. The power button must remain usable while GPIO21 has disabled peripherals.

SOS button - GPIO1

Connect the button between GPIO1 and GND. It is active LOW and uses the internal approximately 45 kohm pull-up for normal operation. The reference firmware behavior is a hold of at least 1200 ms to latch SOS. SOS must not depend on GPS, ML, or a working modem to latch locally.

If SOS wake-from-deep-sleep is later enabled, add a qualified external always-on pull-up because internal pull-ups may not hold during deep sleep.

User button - GPIO2

Connect the button between GPIO2 and GND. It is active LOW and uses the internal pull-up for normal operation. The reference behavior is a short tap for markCount and a long press of at least 2000 ms to clear a locally latched SOS. Clearing a local latch must not erase the server-side event or falsely imply assistance.

9. GPIO21 peripheral rail switch

The switched 3.3 V rail is created by a high-side P-FET or a specifically qualified load-switch implementation:

```
Always-on 3V3 -- P-FET source
P-FET drain -- switched 3V3 -- TFT/touch/MPU/GPS VCC
GPIO21 -- gate drive
Always-on 3V3 -- 100 kohm gate pull-up
```

Reference truth table:

```
GPIO21 state P-FET / switched rail Meaning
LOW ON / 3V3 switched live Peripherals powered
hi-Z with external pull-up OFF / rail disabled Reset/shutdown/deep-sleep-safe default
```

The supplied reference distinguishes this active-low P-FET circuit from a TPS22918. They are not pin/control-equivalent and must not be interchanged without a new schematic and polarity review.

The current firmware uses a driven HIGH for initial software-off and LOW for on; this is electrically consistent with active-low polarity but does not fully implement the reference's hi-Z sleep convention. The external pull-up must still establish the hardware reset default. Rail-off and deep-sleep sequences require bench validation.

10. A7670C cellular modem - reserved in this build

The A7670C is **not wired or enabled in the current build**. Keep the reserved pins unconnected

until the board, regulator, level shifter, PWRKEY, SIM, antenna, and TLS transport are fully qualified.

```
Modem function ESP32 reservation Current state
Modem TX -> ESP RX GPIO48 Reserved; leave unconnected
ESP TX -> modem RX GPIO47 Reserved; leave unconnected
Modem PWRKEY no assigned GPIO Reserved; no code now
Modem VBAT separate 3.4-4.2 V supply Never ESP32 3V3
Modem GND common GND Required return
```

Required future modem power characteristics from the reference:

- Dedicated 3.4-4.2 V regulator/supply rated for at least 2 A.
- At least 300 uF bulk capacitance plus local ceramics.
- Common ground with the ESP32.
- UART interface level-shifted to 3.3 V after verifying the exact module breakout voltage.
- Future PWRKEY operation is approximately a 50 ms LOW pulse, but the actual board must be validated before assigning a GPIO.
- LTE transmit bursts can approach 2 A; the ESP32 3V3 rail must never be used for modem VBAT.

CRC32 can detect accidental transmission corruption, truncation, or mismatch. It is **not authentication or encryption**. Production cellular traffic requires HTTPS/TLS with certificate/hostname validation, device authentication, replay protection, and server authorization.

11. microSD status

The supplied connection reference does not define a microSD chip-select GPIO for this build. Therefore:

- Keep SMARTTRACK_SD_CS_PIN=-1 / disabled until the physical display/SD board and CS net are confirmed.
- Do not reuse GPIO3 or any other sketch placeholder without an approved schematic.
- If enabled later, the SD card must be checked for 3.3 V logic compatibility and added to the SPI CS mutual-exclusion table.
- SD write, power-loss recovery, and offline queue behavior require a real card and fault-injection test.

12. Power budget and rail planning

The reference values are datasheet-typical or community-measured estimates, not measurements on the assembled SmartTrack board.

```
Load Rail Typical reference Peak / note
ESP32-S3 core, Wi-Fi off always-on 3V3 ~120 mA Wi-Fi TX bursts ~283-340 mA
ILI9341 TFT + backlight switched 3V3 ~100 mA ~20 mA logic + ~80 mA backlight
NEO-M8N GPS switched 3V3 ~29 mA 67 mA max; cold acquire ~50 mA
Active GPS antenna GPS module path +10-20 mA Depends on antenna
MPU-9250 switched 3V3 ~3.9-4 mA ~8 uA sleep reference
XPT2046 switched 3V3 <1 mA Negligible reference
```

The normal switched cluster is approximately 133-134 mA before antenna allowance and design headroom. The source document reports about 274 mA total with Wi-Fi off, which includes headroom/antenna allowance. A genuine 800 mA-1 A 3.3 V regulator fed by a 5 V source rated at least 1 A is the conditional reference recommendation. A bare 500 mA USB port is marginal when Wi-Fi bursts overlap the peripheral load.

Required power checks before module connection

1. Check for shorts between every rail and GND with power removed.
2. Confirm always-on 3V3 and switched 3V3 polarity and voltage with a meter.
3. Confirm GPIO21 gate pull-up and P-FET source/drain orientation.
4. Power the ESP32 alone and confirm the switched rail default state.
5. Add GPS/TFT/IMU/touch one at a time while monitoring rail voltage and current.
6. Measure brownout behavior during GPS acquisition and TFT backlight enable.
7. Do not connect the A7670C until its independent regulator and bulk capacitance are load-tested.

13. Staged bring-up and acceptance tests

Stage A - unpowered inspection

- Verify every table row with continuity testing.
- Verify all module grounds join the common ground.
- Verify no ESP32 GPIO is connected to a 5 V signal.
- Verify TFT and touch CS are separate and not shorted.
- Verify GPS TX/RX crossover.
- Verify MPU AD0/NCS/FSYNC straps.
- Verify button pull-ups use the always-on domain.
- Verify A7670C is physically separated from ESP32 3V3 and remains reserved.

Stage B - MCU and rail

- Boot ESP32 with all peripherals disconnected.
- Confirm GPIO21 external pull-up holds the rail off during reset.
- Confirm firmware enables the rail only after safe GPIO setup.
- Measure always-on and switched 3V3 under no-load and incremental loads.

Stage C - sensors

- GPS: confirm 9600-baud NMEA bytes, checksum-valid sentences, then wait for a real fix.
- IMU: scan I2C and verify address 0x68 / expected WHO_AM_I.
- Confirm missing GPS/IMU values remain unavailable rather than being shown as zero.

Stage D - display and touch

- Run TFT reset/fill/text test.
- Run touch test with TFT CS held HIGH.
- Calibrate touch coordinates and document rotation.
- Enable backlight through the NPN stage and measure current.

Stage E - storage and cellular

- microSD remains disabled until its CS is qualified.
- For modem bring-up, use a dedicated supply, level shifters, antenna, SIM, certificate chain, clock, and HTTPS endpoint.
- Test retry/idempotency and confirm records are removed from the offline queue only after an exact server receipt.

Stage F - emergency path

- Hold SOS for at least 1200 ms with GPS unavailable and confirm local latch.
- Confirm the event is queued without ML inference.
- Confirm server delivery, acknowledgment, and resolution are recorded as separate states.
- Confirm the user-button clear action does not erase server audit history.

14. Troubleshooting matrix

Symptom	Likely cause	Corrective action
White/blank TFT	RESET8/DC9/CS10/backlight5 or rail	Check wiring, CS, reset timing, NPN driver, and switched 3V3
TFT works but touch is dead	Touch CS14/MISO13 or SPI mode	Check CS exclusivity, shared MISO, IRQ15/polling, calibration
No GPS bytes	Wrong crossover, baud, VCC off	GPS TX->GPIO6, GPIO7->GPS RX, 9600, switched rail on
GPS bytes but never a fix	Antenna/sky view/acquisition time	Clear sky view, active antenna, wait, add GPS bulk cap
IMU not detected	Address/straps/SDA/SCL/pullups	AD0 GND, NCS 3V3, FSYNC GND, address 0x68, verify 4.7 kohm
Reset during GPS acquire	Rail or regulator sag	Add/verify 100 uF caps, stronger source, measure current
Buttons never register	Pull-up on switched rail	Move pull-up to always-on; verify active LOW
Everything dies after shutdown	Expected switched-rail behavior	Only always-on core/buttons remain; verify wake plan
Wi-Fi causes resets	Weak USB/source transient	Stronger 5 V source and bulk capacitance; duty-cycle Wi-Fi
Modem brownouts	Wrong rail or inadequate supply	Separate 3.4-4.2 V >=2 A supply and >=300 uF bulk
Modem sends plaintext	Unsafe transport fallback	Disable modem; require qualified HTTPS/TLS and certificate validation

15. Project-specific verification findings

The independent review set identified these important alignment items:

1. The hardware reference requires power button GPIO4, but the current firmware configuration and main loop do not yet implement GPIO4 shutdown/wake behavior.
2. The hardware reference assigns MPU INT GPIO16 and static straps; the current firmware only initializes I2C and does not yet provide a qualified MPU driver.
3. TFT/touch GPIO constants exist, but a complete runtime display/touch driver and CS transaction arbitration are not yet proven by hardware tests.
4. The current firmware defaults SD CS and modem PWRKEY to -1, which is the safe choice until those connections are physically qualified.
5. GPIO21 software polarity is active-low correct, but the reference’s hi-Z rail-off behavior and reset/deep-sleep sequence still require board-level validation.
6. Current and capacitor values in this document are estimates from the supplied reference, not certification for the actual ESP32 board revision or breakout modules.

16. Final do-not-connect summary

- Do not connect 5 V to an ESP32 GPIO or 3.3 V-only signal.
- Do not drive TFT backlight current directly from GPIO5.
- Do not pull GPIO10 and GPIO14 LOW together.
- Do not put button pull-ups on switched 3V3.
- Do not power A7670C from ESP32 3V3.
- Do not connect reserved GPIO47/48 or invent a PWRKEY GPIO in this build.
- Do not enable an unknown microSD CS pin.
- Do not treat CRC32 as authentication.
- Do not treat a stale/last-known GPS coordinate as current.
- Do not present an ML Trust Score until a real, evaluated, promoted artifact exists.
- Do not consider this manual a replacement for continuity, voltage, thermal, RF, and end-to-end bench testing.